Is

Relation Without Time, Cause, or Rule

Matthew Parslow
Independent Researcher

Draft — draft-2026-08-30.4
30 August 2026 • document revision 67

DRAFT — draft-2026-08-30.4 r67 (30 August 2026). This document is under active revision and is circulated for comment, not as a finished result. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

One claim, made without machinery. To be is to stand in relation, and nothing further is required for there to be anything at all — no sequence, no cause, no rule about what may follow what. The companion accounts begin after this point and say so; this one supplies what they begin after. It is short because there is very little to say once the claim is stated. The work is not in the claim but in making a case for it.

1 The claim

What is, is. That is the whole of it, and everything below is an attempt to say what the sentence commits to and what it does not.

To be *something* rather than nothing at all is to be distinguishable, and to be distinguishable is to stand apart from something. *Distinction is relation.* There is no third option in which a thing is determinate and stands in no relation whatever: such a thing would differ from nothing, and a thing that differs from nothing is not a thing but the absence of one.

Proposition 1.1 (To be is to be related)**.** *Anything that exists as a determinate something stands in at least one relation. Contrapositively: where no relation holds, there is no distinction, and where there is no distinction there is no determinate something — not one that cannot be told apart from nothing, but no second thing there to be told apart.*

The second sentence was formerly given as *equivalent* to the first and phrased as a distinction *not available to be drawn*, which was a mistake worth correcting in the open, because it made an ontological claim sound like an epistemic one. Availability implies somebody to whom something is available, and there is nobody here: the base of this account has no observers and no views at all, so a criterion resting on what can be told apart would have nothing to rest on. The claim is not that an unrelated existent is one we could never detect. It is that the description picks out nothing — there is no fact it could be a fact about, because a fact requires two things and a relation between them. That reading owes nothing to verificationism, which the account could not use anyway: what exists here does not wait on observation, and a tree falling unobserved is exactly as related to its forest as one that is watched.

What follows in this paper is therefore not an elaboration of the claim but a case for it, and the distinction is worth keeping because it sets what would count as a rebuttal. There is no

complexity in the *claim* to disagree with: the sentence is short enough to hold entire, it says less than any account of what there is has any business saying, and a reader can see all of it at once. That is the intended difficulty, and it is a claim about the proposition rather than about this document — the method, the reading of how formalisms move, and the accounting of what derivation costs are all separable and all rejectable on their own, as the next section says. What the shortness buys is narrower: the proposition cannot be rescued by adjusting an apparatus, because it has none, so the burden on it falls entirely on whether the case is compelling.

The companion accounts begin from this claim and add to it what a world in which things *happen* costs, which this paper is careful to say is not required for there to be anything. An earlier version of this paragraph said they add *exactly one thing*, and that was wrong, and the paragraph's whole purpose was to state something checkable. The companion's own ledger, corrected after audit and moved in both directions since, carries the admission and two postulates named there — a realisation operator and a destroy/preserve distinction — with its Axiom −1 a discipline against importing unpaid assumptions in that paper and a commitment that can fail in the groundings. Transformation is not among them: something has to do the happening, whatever does it is a graph because everything is, and the least graph that can do anything is a pattern together with a change it makes — a rule. So a rule is a graph that does something, and rules running is what this paper's own claim looks like where something occurs rather than a second kind of thing put beside it. That is this paper's method used one level along: it asks what the least structure that can *be* is and answers relation; the companion asks what the least structure that can *do* is and answers rule. The economics does not cost extra either, and the reason is an asymmetry between what this paper claims and what the companions add. What funds a change is another change — the same kind of thing — so the economics admits no second kind of thing, which is category-preservation and not, as an earlier version of this paragraph claimed, closure. The event order does not loop: the companion's own axiom makes enablement asymmetric, so the funding relocates outward rather than closing, exactly as provenance does everywhere else in that account. What a thing does, when it does anything, is *change*; the cost of a change is not a quantity in another currency it is priced in, it is the change; and cost can be unboundedly small and is never zero, with any particular floor belonging to the groundings.

Realisation is a different matter and this paragraph got it wrong twice. An argument that it folds away — nothing selects, what matches writes, two rules matching one state make a bigger change — is recorded in the companion's ledger as attempted three times and withdrawn each time, most recently because it trades collapse for union and because residual cannot carry an incompatibility between two individually admissible writes. It is a postulate there, and this paper should not have said otherwise.

So what the companions add to this paper is the admission that things happen, and past that, the bets they state as hypotheses rather than as commitments. What those commit to together is then unpacked at length, and the unpacking is not further claims about what exists.

The strong form of the statement is therefore false and the useful form survives: the base is small, what is added to it is enumerable, and anything in the companions that is an independent posit uncounted by that enumeration is a defect in their economy. The count has moved repeatedly under that standard — five posits found where two were claimed, folds made and two of them found unsound and reversed, claims narrowed where they had been stated too strongly — and the movement is the procedure running rather than a caveat on its result. A ledger that had never moved would be evidence of nothing except that nobody had checked it. What should be read from the destination is an increase in exposure and not a reduction: an account resting on fewer things has fewer places where a failure would be contained.

The relation between this paper's length and the companions' runs in the direction one might not expect. This paper's shortness is not a contrast with the length of the companions but the thing that licenses it. A reader meeting a fifty-page theory without this one in front of them has no way to read its length except as the size of what it assumes, and an account that assumed

fifty pages' worth would be an extravagant thing to be asked to grant. Put the base first and the same fifty pages change character: they are the length of a *derivation*, not of a commitment, and the two are not costs of the same kind. This is the ordinary accounting of minimum description length applied to an account rather than to a string — what is charged for is the generating structure and not what unrolling it produces, so a short generator with a long output is cheap and a long generator is not, however short its output. The companions are meant to be output.

Meant to be, and not wholly: where a companion introduces a *definition* — an aperture, a change grammar, a residual, a frame — it is adding vocabulary, and vocabulary sits on the generator's side of that ledger however faithfully it was motivated. Only the reasoning *over* those terms is output. So the licence covers less than all of the length, the fraction it covers is not measured anywhere, and the accurate statement is a comparison rather than an exemption: fifty pages of derivation over a declared vocabulary is a smaller commitment than fifty pages of assumption, which is not the same as costing nothing.

Which is also the standard this paper puts itself under, since the reasoning is comfortable enough to be dangerous. *It is derivation* is available to be said about any quantity of text whatever, and saying it costs nothing; what stops it being an excuse for length is that each part must be traceable to the base and the tracing must be something a reader can attempt and fail at. Where that is not available, the licence is not either, and the length reverts to being what it appeared to be.

2 Which is to say: a graph

The claim has a name, and refusing to use it would be coy. Relata standing in relations are nodes standing in edges. *Everything is a graph* — and that sentence is the whole thesis of the companion accounts, stated here in the only form that requires nothing to state it.

The question this paper asks is the dumbest one available: *what is there?* A child can ask it and nothing prevents anyone answering. It is also the largest, because an answer to it constrains every answer to anything else, and because there is nowhere further back to stand.

Two things follow about the form of this paper. It builds *no machinery*, deliberately: no transformation, no grammar, no economics — none of the apparatus a reader could dispute *instead of* disputing the claim. That is not the same as offering nothing else to disagree with, and an earlier version of this paragraph said it was. What follows below is a method, a reading of how formalisms move under compression, and an accounting of what derivation costs; each is separable from the claim and each can be rejected on its own. The honest form is narrower: no *structure is posited* here that could absorb an objection meant for the claim, so a reader who rejects the claim cannot be answered by adjusting a mechanism, because there is none to adjust.

And the objection that arrives immediately can be answered in a line. *Everything can be represented as a graph, so the claim is empty* — yes to the first clause, and the answer is that this is not a claim about representability. Nodes and edges are cheap; what is asserted is that there is *nothing else*, that relata and relations exhaust what there is, and that time and cause and rule in particular are not among the furniture. And the assertion is about *this* instance rather than about what is possible: a world with a global clock in its floor, or with production as a primitive, is not prohibited by anything here — such a world could be built and would be perfectly coherent — the claim is that *this one is not that.* What would bear on it is therefore a demonstration that this world has such a floor, which is an empirical matter and not a modal one, and imagining a world that does costs nothing and settles nothing.

That is also why finding one would bound the account rather than refute it, in the way a later theory bounds an earlier: the claim would hold of worlds without such a floor and this would not be one of them. The companions argue all of this at length; here it is a paragraph, because here there is nothing else to argue about.

3 How the claim is reached

A claim of this shape is worth nothing if it arrives by assertion, and the difference between an account and a pronouncement is whether there is a procedure behind it that someone else can run. There is one here, and it is short.

Ask at maximum scope. The question is put to everything at once rather than to a domain, because a shortest sufficient account of one domain tells you about that domain and a shortest sufficient account of all of it tells you what is common to all of it. The scope is what does the work: the more that must be accounted for, the more the structure peculiar to any part washes out and the more only what is shared survives.

Prefer the shortest account that suffices. Between two accounts covering the same ground, the one carrying less structure is preferred, and this is the ordinary discipline of minimum description length rather than anything invented here. Applied at maximum scope it has a direction: keep removing until removing costs coverage.

Remove, and see what breaks. This is the operation and it is the only one. Take a candidate — time, cause, rule, substance, an observer, a global clock — and put the account without it. If what was accounted for is still accounted for, the candidate was not required and it goes. If coverage is lost, it stays, and the loss shows what it was doing. Each removal is a test whose outcome is determinate in one direction and not in the other, and the asymmetry is this: a *successful* removal settles the matter, since an account that stands without the candidate is exhibited and can be inspected, while a *failed* one records only that nobody has yet seen how, and stays open to a reader who does. That is what makes the procedure repeatable rather than a matter of taste, and it is also why the paper expects to be shortened.

Run that far enough and it stops in one place. Time can go: nothing in the claim requires sequence. Cause can go: relation is not production. Rule can go: nothing needs to be permitted or forbidden for things to stand as they stand. Relation cannot go, because removing it removes distinction, and removing distinction removes the possibility of there being determinate somethings for the account to be about at all. *The claim is not asserted here. It is what survived.*

A removal must be named and restorable. This is the discipline that separates the method from mere simplification, and it is the one place the procedure can be abused. Every account reached this way is an idealisation, and an idealisation is just a frame — which is no defence on its own, since everything is one. What makes a frame cheap to hold rather than expensive is whether what it removed is written down and can be put back. The rock is what this claim looks like at the grain at which it is a rock; that its atoms move and it weathers at other grains belongs on the same page, so the removed structure is recoverable and the idealisation costs little. *The spherical cow is not objectionable for being spherical.* It is objectionable when nobody writes down that it was made spherical, and the result is then applied to a cow.

The preference for the shortest sufficient account is not this programme's invention. It is Rissanen (1978)'s: do not introduce structure that buys no additional distinction. The information-theoretic setting it is stated against is Shannon (1948), which supplies the measure and the coding bound and does *not* supply the parsimony norm — it contains no model selection and treats redundancy as useful where a channel is noisy, so citing the two together for the norm, as an earlier version of this sentence did, borrowed a name that did not make the claim. What is added here is only the scope at which the norm is applied.

Two honest limits on the method, and the second matters more.

Removal establishes that nothing *tried* was required, which is not the same as nothing being required. The result is only as good as the candidate list, so that list should be stated rather than implied: time, causation, substance or intrinsic nature, an observer, a privileged frame or global clock, admissibility, and — less obviously — any axiom restricting where a distinction may be made from. That last is worth naming because it is easy to keep by habit: a rule that every distinction is made in some frame governs frames, and there are none here to govern. What

removing it does *not* do is remove every commitment about distinction, and the difference is worth being exact about since the two are easily run together. Removed: any rule about where a distinction may be made *from*. Retained, and load-bearing: that a distinction is a relation, which is the identification the whole claim turns on and which is a commitment rather than a discovery. A reader who holds that a thing can be determinate without standing apart from anything rejects this paper at that sentence and needs to go no further — which is where the disagreement should be, rather than several pages downstream. It arrives with the machinery it constrains, in the companion account, and is not required for anything this paper says. Anyone who thinks something else belongs at the base is invited to add it and run the same procedure, and that is the form disagreement with this paper should take.

And the procedure does not deliver certainty, only survival. A candidate that has not yet been removed successfully may still be removable by someone who sees how, and the account would then be shorter than it is here.

One operation, applied to what came bundled

It should be said plainly that there is only one move here and that it is used repeatedly, because a reader who expects a method proportioned to the size of the claim will go looking for machinery that does not exist. The operation above is stated as removing posited structure — time, cause, rule — but its more productive target is different, and is the same operation pointed the other way.

Almost every claim worth taking at maximum scope is one that some field already accepts, and accepts with a restriction attached. Relations are basic *in this formalism*. There is no privileged observer *in mechanics*. The shortest description is preferred *for this class of model*. The restriction is generally not a discovered property of the subject; it is the restriction that constitutes the field, and it travelled with the claim because within a field there was never an occasion to separate them. So the move is to take the claim seriously and decline the restriction — which is not a new procedure but *remove, and see what breaks* applied to an inherited limit rather than to a posited entity, with the same asymmetric outcome and the same requirement that the removal be named and restorable.

Two things follow. The first is that results reached this way look simultaneously trivial and consequential, and are both: trivial because nothing was added and the claim was already granted, consequential because the discarded restriction was doing work that nobody had audited, having never been separable from the subject inside the field that held it.

The second is a limit, and it is severe. Most restrictions are load-bearing — they are attached because removing them breaks something, and taking a claim past its restriction usually produces nonsense rather than generality. The operation is cheap to run and returns nothing on almost every attempt — a rate this paper commits to in Section 9 and cannot yet support, since no ledger of the failures was kept and the base rate is therefore reconstructed rather than recorded — and the ones that return something are worth reporting for that reason and not because the operation is powerful. Nothing here is a recipe for producing results by taking things literally. It is a cheap test, run often, with a low yield, and a signal on firing that is clear about *that* something did not break while saying nothing about whether what survived is worth having.

4 A consequence with a track record

The claim fixes what compression is aiming at, and that turns out to be checkable in a way the claim itself is not.

If what there is, is relata standing in relations, then the shortest sufficient description of anything is stated *in terms of* relata and relations. The qualification is doing real work and an earlier version of this passage omitted it, giving instead that a description shorter than its

subject's structure has lost something — which is a picture theory, and worse, contradicts the accounting two sections above. Minimum description length charges for a generator, not for what unrolling it produces: the shortest sufficient description of the complete graph on a million vertices is a phrase naming the relation and the count, not five hundred billion enumerated pairs, and a description isomorphic to its subject is very often the expensive one. So the claim cannot be that the shortest description *enumerates* relata and relations.

What it can be, and is, is a claim about the vocabulary the generator has to be written in. "The complete graph on a million vertices" is a short generator whose terms are a relation type, an arity and a count; it is about relata and relations without listing any. The prediction is therefore that formalisms under compression converge on *primitives* that are relata and relations, not on descriptions the size of their subjects. *The ideal information shape is a graph* in that sense — what the terms denote is fixed by what is being described, while how briefly they can be combined is not. So a field pushing its own formalism toward minimum description should *move toward* graph form, and it should do so without anyone intending it.

That is a prediction about the history of formalisms rather than about the world, and it appears to hold: physics reaching for spin networks, tensor networks and causal sets; chemistry settling on the molecular graph; linguistics on dependency structure; computing on data structures that are graphs and on a categorical foundation that is arrows before objects; neuroscience on the connectome; biology on phylogenies and interaction networks. None of these was adopted because anybody held the thesis of this paper. Each was adopted because it was the cheapest thing that still accounted for the subject.

One case is worth setting out rather than listing, because it is the oldest, it has been kept longest, and it is usually described as an invention. *Double-entry bookkeeping is not an invention. It is a compression.* A transfer has two sides, and the record that keeps both of them names two accounts and the movement between them — relata and a relation and nothing else. It should be said at once that this is not a minimality claim, because Ellerman refutes the obvious one: a single-sided system using signed numbers records the same information, and given the same opening equation and the same transactions the two end in the same place. What single-sided recording loses is the trial balance, and that is the whole of the difference worth having here. Nobody derived it from a thesis about what exists.

The history is what makes it a case rather than an analogy. Pacioli (1494) is usually credited with the system, and what he wrote was a description: the *Particularis de Computis et Scripturis*, a treatise inside a mathematics textbook, setting out a method already in use among Venetian merchants. The practice is older than the account of it by roughly two centuries — Lee (1977) reads the Farolfi ledger of 1299–1300 as full double entry — so the system was found in use and then written down, which is the order a compression arrives in and not the order an invention does. And it has been kept for five centuries, in the formalism the field teaches and computes with, which is what the commitment below actually asks for.

The honest counterweight belongs here too, because the history is contested in a way that bears on the claim. Yamey (1949) argued against the then-standard view that double entry was adopted for rational profit calculation, holding that early merchants did not use it that way. What Yamey found, though, is a *smaller* purpose rather than none — tracking debtor and creditor relations, custody, and agency accountability — and Sangster (2016) names it more precisely still: the system gave a clear picture of every debtor and creditor *together with the means to check that the entries between them were complete and accurate.* An earlier version of this paragraph inferred from Yamey that the method was adopted with no theory of what it was for, and that inference does not survive: it was adopted for something, and the something was checking. That is worse for one argument and better for the next, and both are stated rather than the convenient one alone. It removes any claim that nobody designed the technique for anything. It also means the error-detection property below was not a lucky side effect — it is what the thing was built to do, which is why it was kept.

And it earns the case rather than illustrating it, because of what the practice detects. A ledger that does not balance is not a rule violation but a *residual* — something recorded that does not cohere with the rest — and the trial balance is a coherence check, which is why the technique catches errors at all. A convention that did not match the two-sidedness of what it records would have no such property, so the survival of the technique is evidence about the subject and not only about the bookkeepers.

That reading has a mathematical statement and it is not this paper's. Ellerman (1985, 2014) shows that double entry implicitly uses the *group of differences* over pairs of unsigned numbers — the T-account is the pair — so a transaction is a difference and the two entries are its two components rather than one record written twice. Balance is then the group identity, and an unbalanced ledger is a residual in the exact sense, which is where the error-detecting property comes from. The dating is the part that bears on the claim here: the construction was only formulated abstractly in mathematics in the nineteenth century, after the practice had used it for over five centuries. So the structure was in use long before anyone had the mathematics to say what it was — found, then written down as a method, then formalised four hundred years after that. Arya et al. (2000) take the same structure the other way, recovering transactions from statements by treating accounting information as a linear aggregation whose rule is double entry; and Arya et al. (2000b) give the reading this paper actually wants, characterising the consistent transaction vectors by a *directed graph representation of the accounting system* without which the inversion becomes intractable as accounts multiply. That is the relata-and-relations reading arrived at without the thesis. The graph framing is the companion paper's rather than the estimation paper's, and citing only the second for it — as an earlier version did — attributed to one paper what the other says.

The discriminator matters and is easy to lose. *The prediction is about movement, not about drawability.* That a field's subject matter can be drawn as a graph shows nothing, since graphs are expressive and almost anything can be. What would count is a field that compressed its formalism and moved *away* from graph form — toward a primitive that is not a relatum or a relation and cannot be reduced to them — and that is what the second commitment below asks for.

5 One is not enough

A single relatum with nothing to relate to is not a minimal case of existence but a failed one. There is nothing for it to be distinguished from, so there is no fact about what it is; it does not sit at the bottom of the hierarchy of existing things, it fails to enter. *One relatum is decoherent.*

The empty case is the other end and is cleaner. Nothing at all stands in no relation, has no parts to be in tension, and admits no failure of fit — it is complete in the only sense available to it. That is why the empty set is the only perfect thing and why perfection is not something an existent can have: *to exist is to be imperfect*, because to exist is to be related, and to be related is to be partial with respect to what one is related to.

6 One direction only

A consequence of the restraint deserves its own statement, because it is what makes the claim's subject matter peculiar. *What merely is cannot observe; it can only be observed.*

To observe is to make a distinction, and making one is an act — something is done, and afterwards the observer holds what it did not hold before. That requires transformation and some grammar of what may follow what, and this paper has neither. So what is described here does nothing at all, and observing is among the things it does not do.

It can be the *object* of observation without difficulty, and in practice it is nothing else. A rock is what this looks like from outside: met, related to, distinguished by something that does

have the machinery. But there is no corresponding inside. *There is no view from anywhere* — not a narrow view, not a limited one, none.

The reason is not that framehood needs rules — it does not, and the companion is explicit that a structure with no rules of its own is still a frame, because *individuation is performed by the container and not by the thing.* A boundary belongs to the frame that draws it, so what a rule-less structure lacks is not framehood but the capacity to individuate anything itself. It is also why an *empty* frame is impossible rather than merely unattested: an edge needs relata, so where there is nothing there is nothing to demarcate.

What the base lacks is the drawing. A frame is a boundary relation *instantiated*, and instantiating is something a container does; a level at which nothing runs has nothing doing it, so it has no frames — not because a frame would need rules of its own, but because it needs someone else's.

What is missing is *making.* A view is not had by standing in a relation; it is had by a frame that *makes* a distinction, and making is doing, and doing is what this level does not have — that is precisely what the companions add when they admit that things happen. So the familiar claim, that every view is had from somewhere by someone positioned, presupposes there are views and asks where they are from. Here there are none to ask about: *no view from nowhere, and no view from anywhere either*, not for want of a place to view from but for want of anything doing the viewing.

An axiom restricting where a distinction may be *made* from therefore cannot be stated at this level, let alone assumed — because it governs an act, and there are no acts here — which is also why there are no frames at this level to govern, since instantiating a boundary is one. It arrives with the machinery that performs them.

Proposition 6.1 (Observation is one-directional at the base)**.** *What stands in relation without transformation or grammar can be observed and cannot observe. It is therefore always the object of a distinction and never the maker of one.*

7 What is not claimed

The claim above is smaller than it may appear, and the restraint is the point.

No time. Nothing here requires sequence. Relations hold; they do not happen. There is no earlier and no later in this paper, and no order in which anything is arrived at.

No cause. Nothing here makes anything else be. Relation is not production, and the account has no resources with which to say that one relatum brings about another.

No rule. Nothing here forbids anything or licenses anything. There is no admissibility, nothing that may or may not follow, and therefore nothing that could be violated.

Those three are not oversights and they are not deferred out of modesty. *They cost extra.* A world in which things also *happen* requires transformation and a grammar of what may follow what, and that is what the companion accounts introduce and pay for. This paper is what remains when they are removed, and it is offered as evidence that they are separable rather than constitutive.

8 Prior positions

None of this is new and it should not be dressed as though it were. The neighbours are easy to name. That relations are prior to, or on a level with, their relata is the ontic structural realist position (Ladyman and Ross, 2007; Esfeld, 2004); that a thing is nothing over and above what it stands in is older and starker (Nāgārjuna, 1995); and the bundle tradition reaches a neighbouring conclusion by a different route (Williams, 1953).

The relation to them is not the usual one, and the difference can be stated exactly rather than as a matter of emphasis. *Each of those accounts supposes a particular* ***type*** *of relation where the argument only requires that there be one.* Ontic structural realism is a thesis about physics — about what our best theories say the world is made of — so the relation carrying its weight is measurement: an observer edge, running from something that theorises to something theorised about. The perspective literature does the same with an edge of viewing. Neither is mistaken about the structure; both are over-specified about the edge.

The removal is immediate and is the method's own. Take the observer edge out and put any other edge in its place — a gravitational one, a chemical bond, mere adjacency, one part of a rock against another — and everything the structural claim needs is still there. Two relata and a relation between them do not care which relation it is. So what those accounts treat as constitutive is a *choice of edge type*, salient in the domains they were written for and doing no work in the claim itself; and what survives removing it is the claim this paper makes.

The same over-specification is visible in the two fields that own this vocabulary, which is worth saying because they are the ones a reader will expect to have settled the matter. Information theory defines its quantities over a source and a channel: there is something sending, something receiving, and a distribution over what might have been sent. That is an observer edge again, wearing different clothes, and it is why entropy is a property of a source-and-receiver arrangement rather than of a thing. And graph theory, in its standard presentation, takes a vertex set first and defines edges as pairs drawn from it — which builds in the priority of relata over relations before any question about that priority has been asked. Neither field is careless. Each fixed an edge type or an order of construction because its own problems required one.

Which yields the deflationary version of this whole paper. *If everything is a graph, then the minimum structure required to be a graph is the minimum structure required, full stop.* That structure is not something this paper discovers: it has been written down in every textbook of the subject for a century, as relata and relations and nothing else. What is being done here is not deriving it but *taking it seriously* — reading a definition that was built as a formalism as though it were an answer to the largest question, and then checking that nothing else has to be added. The removals above are that check. That is the entire contribution, and it is smaller than it sounds while being, if the claim holds, sufficient.

One qualification even there, and it is not a criticism of the formalism. The textbook definition takes a vertex set as given and edges as pairs over it, which makes relata prior — and it does so for a good reason. *Graph theory's objects do not need to exist.* They are built, and building takes an order: you must have the vertices in hand before you can say which pairs of them are edges. The priority is a feature of *construction*, and construction is what mathematics does.

Nothing here is constructed. What exists was not assembled in an order, and there is no stage at which the relata were available and the relations not yet drawn. So the priority does not merely go unneeded in this paper — it is *unavailable*, and for a reason already established: an order of construction is a sequence, and there is no time here for one to run in. A definition that builds cannot be read as a description of something that was never built without carrying its scaffolding across.

Which sharpens the earlier point rather than softening it. Graph theory is not over-specified for its own purposes; it is exactly specified for constructing objects. What does not transfer is the scaffolding, and noticing that it is scaffolding is most of the work of taking the definition seriously.

That is also why the closest neighbour is the one furthest from physics. Nāgārjuna (1995) makes the claim in a form that does not require an observer edge — nothing needs to be looking, measuring or theorising for a thing to be nothing over and above what it stands in — which is the property the others lack, and which only became the discriminating one once the edge type was seen to be removable. That reading is contested and the stronger form given here

previously, that there is no observer edge anywhere in the account, is not one the secondary literature licenses: conceptual or nominal dependence is standardly the third and strongest of the dependence relations at issue, and on that reading the carving is ours. What the comparison needs is only the weaker claim, that the position is *statable* without the edge, which both readings allow.

There remains the fact that this is simple enough to state in a sentence and appears not to have been stated, which invites the reaction that it must therefore be either trivial or wrong. The account has its own explanation for that pairing and it is in the companion on assessment, so a short one will do here, and the short one is structural rather than a remark about incentives. *People do not study everything. They study something.* A field is constituted by its restriction — what it attends to and what it sets aside — and a field with no restriction is not a field. So the question this paper asks, which has to be put at maximum scope or it is not the same question, belongs to no discipline: there is no subject matter of which it is the subject matter, and no method calibrated for it, because every method is calibrated to a something.

That is why the general form goes unwritten while every local version gets written well.

And the mechanism has a worked case in this paper already, which is worth pointing at because it makes the explanation checkable instead of merely available. The gap between double entry being in daily use and the group of differences being written down is about three and a half centuries — Pacioli in 1494 to Hamilton in 1837, on Ellerman's own attribution — on top of the two centuries between the practice and Pacioli's description of it. That is not a hard construction: it is how the integers are built from the naturals.

The reason it went unwritten is *not* that nobody looked, and an earlier version of this passage said it was. Mathematicians did look — Ellerman names De Morgan in 1869 and Cayley, who published a pamphlet on double-entry bookkeeping in 1894 — and did not make the ordered-pairs identification. So the case is weaker than a clean two-restriction gap and is still a case: a trivial structure in continuous use by thousands of people, examined by capable people from the other side, went unformalised anyway. Ellerman adds the part that bears on the mechanism, which is that placing the identification took him three decades to publish in an accounting journal. The restriction is real; what it obstructed was the crossing rather than the looking.

The parallel is in the sociology and not in the content, and the difference matters. What double entry says is not what this paper says. What the case establishes is that *a thing can be simple, in constant use, and unwritten for centuries, for no reason except that no field owned it* — which is the mechanism this section claims, evidenced somewhere it can be dated. Whether this paper's claim is another instance of it is not something the case decides, and a reader who thinks the sentence is trivial and wrong rather than trivial and unwritten is disagreeing with the claim, which is the right place to disagree with it. It also explains, without needing a separate reason, why such statements tend to arrive from someone crossing between domains: crossing is what asking at maximum scope looks like from inside an arrangement where every position is a restricted one.

The claim of absence should be held loosely, though, and more loosely than the rest of the paper. It is the kind of claim this programme has repeatedly got wrong, always in the same direction — the prior art nearer than expected — and the sentence above is one nobody could be confident of having searched exhaustively. It is offered as an invitation: if this has been written down as a claim about what exists, rather than as a formalism or as a thesis within one field, the right response to this paper is the reference.

What is contributed is accordingly not the thesis but its *position*: that it is enough on its own; that time, cause, rule, an observer, and any axiom about where distinctions may be made from are not required for it; and that an account of how things change can be built on top without folding any of them back in. The companion papers are that account, and their first move is to assume this one.

9 Falsifiability

Three commitments. The first is what the paper claims; the second is what the claim predicts and is the easiest to test; the third is a claim the paper makes about its own method.

1. [SHAPE] **To be is to be related.** A claim about this instance and not about every possible one. Fails on the exhibition of an extant something standing in no relation whatever — not merely one whose relations are unknown or unmeasured, which is the ordinary condition of most things, but one for which the claim that it stands in any relation at all is false. The burden is entirely on that side: no enumeration establishes the claim, since the domain does not terminate, and one instance would bound it.

   An objection to this commitment has to be met, and meeting it says what kind of claim the proposition is. The objection runs: the paragraph above holds that the description of an unrelated existent *picks out nothing*, so the falsifying case is one in which there is no fact to report, and a falsifier nothing could answer to is not a falsifier. The first half is granted. The conclusion is not, because it assumes that a falsifier which cannot be *coherently* met is thereby not a falsifier, and the distinction those two run together is the one this whole account is about.

   The commitment is falsifiable and it is not coherently falsifiable, and the second is the content of the first rather than a defect in it. A claim with no content has no attemptable counterexample at all: put one to a vocabulary and you get another expression in the vocabulary, which is not a failure of anything. Put one to this claim and the attempt has a determinate outcome — it decoheres, in the companion's exact sense, returning nothing with the whole of the frame as residual. That outcome is checkable, and it is distinguishable by inspection from the vocabulary case: *I tried and got nothing back* is a different result from *there was nothing to try*. What the objection has found is not that the claim risks nothing. It is that the claim is at maximum scope, so the only thing that could refute it is the one thing the claim says cannot be assembled — and a question whose refutation would have to be incoherent is the most a question of this shape can be, not the least.

2. [SHAPE] **Compression moves toward graph form.** A consequence of the first and easier to test. Fails if a field, pushing its own formalism toward the shortest account that still suffices, moves *away* from relata-and-relations — adopting under compression pressure a primitive that is neither, and *keeping* it — carrying it in the formalism the field teaches and computes with rather than in a reduction nobody uses. What is asked is retention and not irreducibility, because graphs are expressive enough that almost any candidate can be argued to reduce, and a test whose confirming door is shut by that fact must not have its refuting door bolted by it too.

   Read strictly, that is already failed, and the failure is visible in this paper's own worked example. *The complete graph on a million vertices* keeps an arity and a count, and neither is a relatum or a relation; every confirming case above keeps something similar — amplitudes, irrep labels, branch lengths, edge weights — and each is taught and computed with. So the commitment cannot be about whether a formalism carries *only* relata and relations, since none does and none is likely to. What it can be about, and is here restricted to, is the *bearer of the field's distinctions*: whether what a field individuates and quantifies over under compression pressure is relata and relations, with the rest carried as labels on them, or whether the individuated thing is something else. A field whose primitive bearer of distinction stopped being relata-and-relations would fail this. That is narrower than the sentence it replaces and it is what the evidence offered actually supports. It also still lacks an operationalisation of *compression pressure*, which is the remaining hole. Expressiveness is not at issue in the other direction either: that a subject *can* be drawn as a graph is no evidence, and only the direction of movement under compression counts.

3. [SHAPE] **The method's yield is low.** The claim that the operation of declining an inherited restriction returns nothing on almost every attempt, which this paper leans on — it is what makes a rare result evidence of a discipline rather than of a machine for producing results. It is also a base rate over attempts nobody recorded, which is a defect and is stated as one. It fails if a record of such attempts, kept prospectively rather than reconstructed, shows the operation succeeding often; and this paper owes that record before the claim should be granted.

One objection to the first of these has to be met, because an earlier version of this section raised it and was wrong. It runs: exhibiting a counterexample means observing it, observing relates it to the observer, so anything offered as unrelated is related by the act of being offered, and the falsifier cannot be met.

That slides from *relation* to *observation*, and the two are not the same. *Observation is not required for existence.* The tree falls in the empty forest and stands in relation throughout — to the ground it strikes, to the air it moves, to its own fibres as they part — and not one of those relations waits on anybody looking. Observation is a *kind* of relation, and a late and expensive one, since it requires a frame with the machinery to make a distinction; relation is the general case and requires nothing.

So the claim here is not that to be is to be perceived, which is a different thesis with a different name and is not held. What is claimed is that to be is to be *related*, and the falsifier asks for something standing in no relation whatever — not for something unobserved, of which there is a great deal. That makes the commitment harder to meet rather than easier, since a candidate cannot be rescued by pointing out that it is now related to whoever produced it: the question is whether it stood in any relation at all, and the fallen tree settles that without an observer anywhere.

Proposition 6.1 survives this unchanged. What merely is still cannot observe. It simply does not follow that it needs to be observed in order to be.

A second thing would count against the paper without falsifying the proposition, and is worth stating separately: if time, cause, or rule turned out not to be separable — if the account of change built on this one could not be constructed without folding one of them back into the base — then the restraint claimed in Section 7 would be a fiction and this paper would be an incomplete statement of the base rather than a minimal one.

References

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

Anil Arya, John C. Fellingham, Jonathan C. Glover, Douglas A. Schroeder, and Gilbert Strang. Inferring transactions from financial statements. *Contemporary Accounting Research*, 17(3):366–385, 2000.

David P. Ellerman. The mathematics of double entry bookkeeping. *Mathematics Magazine*, 58(4):226–233, 1985.

David Ellerman. On double-entry bookkeeping: the mathematical treatment. *Accounting Education*, 23(5):483–501, 2014.

Anil Arya, John C. Fellingham, and Douglas A. Schroeder. Estimating transactions given balance sheets and an income statement. *Issues in Accounting Education*, 15(3):393–411, 2000.

Geoffrey A. Lee. The coming of age of double entry: the Giovanni Farolfi ledger of 1299–1300. *Accounting Historians Journal*, 4(2):79–96, 1977.

Luca Pacioli. *Summa de Arithmetica, Geometria, Proportioni et Proportionalità.* Venice, 1494. The bookkeeping treatise is the *Particularis de Computis et Scripturis.*

Jorma Rissanen. Modeling by shortest data description. *Automatica*, 14:465–471, 1978.

Michael Esfeld. Quantum entanglement and a metaphysics of relations. *Studies in History and Philosophy of Modern Physics*, 35(4):601–617, 2004.

James Ladyman and Don Ross. *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford, 2007.

Nāgārjuna. *The Fundamental Wisdom of the Middle Way: Nāgārjuna's Mūlamadhyamakakārikā*. Translated by Jay L. Garfield. Oxford University Press, New York, 1995.

Alan Sangster. The genesis of double entry bookkeeping. *The Accounting Review*, 91(1):299–315, 2016. Published online 2015.

Basil S. Yamey. Scientific bookkeeping and the rise of capitalism. *The Economic History Review*, 1(2/3), 1949.

Donald C. Williams. On the elements of being: I. *The Review of Metaphysics*, 7(1):3–18, 1953.